DATA MINING PROJECT

Master in Data Science and Advanced Analytics

NOVA Information Management School

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Amazing Airlines Customer Analysis

Group 29

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1. Abstract

In order to effectively gather a loyal customer base, it is in an airline company's best interest to create a segmentation strategy that involves little resources, while also being efficient in creating groups of similarly behaving customers. The vast amount of data they have access to provides an opportunity to improve the understanding of customer behavior and travel routines. In this project, we address the first step toward identifying the most relevant customer characteristics, which will serve as the foundation for a complete customer profiling process. By analyzing the datasets, engineering new features, and integrating external data, we created enriched datasets capable of offering deeper insights into the customer base. Finally, using CRISP-DM methodology, we identify three key insights of the provided data. First 47% of flights are concentrated in three hub cities, second customer lifetime value has stagnated despite 15% program growth, and third current loyalty programs show no behavioral differentiation. We propose a multi-perspective segmentation approach combining RFM analysis, redemption behavior, and geographic patterns to segment customer travel behavior in a clear and practical manner, ultimately supporting performance improvement and strategic decision-making for the company.

2. Introduction

Operating profitably in the Airline industry is notoriously challenging, with McKinsey reporting that 46% of the airlines they surveyed in 2023 were profitable and 41% in 2024 (McKinsey & Company, 2025). Apart from the widely spread challenges caused by supply chain disruption and economic uncertainty, traditional airlines also need to face new competitors entering a market with historically high entry barriers. Competitors such as Ryanair (Europe) or Scoot (Southeast Asia) provide customers with a low-cost, minimalist service that attracts especially hyper price-sensitive customers. This puts more traditional airlines in a challenging position, as they have to attend to their existing customers' needs while also expanding their offerings to more price-sensitive customer groups. One example of addressing this challenge is American Airlines, which introduced up to 5 seating options tailored to its existing and expanding customer segments (AVRAM, 2019). These changes show a clear need for AIAI to revisit its existing three-tier loyalty program (Aurora, Nova, Star). As the current approach lacks clear separation criteria, it fails to distinguish between high-spending business travelers and cost-conscious frequent flyers. Second, the current approach gives only limited consideration for geographic patterns, redemption behaviors, or companion travel behaviors—all factors that could inform targeted service offerings. Finally, as our EDA reveals, the existing loyalty tiers show no clear and separable criteria for customer spending behavior or demographic information. A data-driven, multi-perspective segmentation approach is therefore essential for AIAI to improve its customer understanding, create personalized marketing campaigns, and maintain competitiveness in an increasingly challenging market. For this goal our project aims to identify 3-5 distinct customer clusters with silhouette scores between 50% and 70%. Additionally these clusters should be clearly interpretable from a business perspective, categorising them, for example into, Frequent-Flyer, Best-Customer, Family Traveler or alternative groups which can be clearly separated in a business context.

3. Data Analysis

1. Raw dataset analysis

For the analysis of the customer two datasets were provided (with an additional CSV for the metadata). The first dataset is called CustomerDB and is composed of 16921 instances of clients with 20 columns regarding their anagraphical and financial information and their relationship with the company. The second dataset is called FlightsDB and is composed of 608436 instances of flights with 10 columns describing them with an aggregation on months and years (from January 2019 to December 2021). Each customer has a related month with a summary of the total flights taken, the average length, and information about the points obtained and redeemed in the month. In Appendix A we have an overview of each column's number of missing values which we later address. The complete dataset is obtained by merging the two databases via the Loyalty# column. The relation is a one-to-many from the CustomerDB to the FightsDB. Each customer is related to 36 rows (minus the 20 removed for not having flight records), one for every month in the three years period related to the records. This setting reduces the possibility of recording day specific patterns but can highlight distributions over monthly trends.

One of the first steps in the preliminary analyses was about understanding the quality of the data. Since the customers are connected to the flights via a unique identifier, we wanted to check if that identifier was effectively unique. We found 327 duplicated IDs and decided to drop them in both datasets since a segregation of the flights would not have been reliable enough for our purpose. Another discovery was that all the number of flights with companions and the number of flights of the months in the year 2019 were multiple of 0.9; we assumed an error with the records and we divided them by that value.

Checking for missing values showed us that three columns in the CustomerDB had missing values (zero in the FlightsDB). The columns were Income (20 rows), Customer Lifetime Value (20 rows) and Cancellation date. While the latter was there to symbolize still active subscriptions the other two were more problematic so the choice was to drop 20. We found that they were the same rows and that they had no recorded flights so the impact was minimal on the overall quantity of the data.. Other problems that we noticed came from irregularities with months with zero flights and a non-zero travelled distance, subscription duration with less than zero days, and impossible travel distance. All those observations were dropped.

The last problem was found from the column regarding the salary of the customers. For every customer, whose highest level of education was college, the value of the income column was 0. We thought that it was a problem and that we couldn't drop one fourth of the dataset. We decided to impute it using real world data (External Features integration Section).

Regarding the Duplicate IDs in the CustomerDB and the FlightDB we decided to remove those instances from our datasets. The reason for this is that with the given data we couldn’t reliably map those cases from the CustomerDB to the matching entry in the FlightDB. Additionally this issue only covers a limited subset of our data and thus will only have a limited impact on our analysis.

4. Results

1. Feature engineering Strategy

Given the nature of the datasets we couldn't gather meaningful insights from the raw version of both datasets. Starting with the subscription we highlighted which customers cancelled their subscription and how many days those subscriptions lasted. After that, via merging, we counted how many flights each customer had, how many of those were inside the subscription period, and the relative percentage of them. From those we also calculated how many flights with or without a companion were done inside the subscription period. Other insights regard the proportion of flights taken with a companion.

For financial insights we divided the lifetime value by the number of flights to get the mean amount of money spent by each customer per flight. We followed by getting the total points earned and redeemed. We calculated the number of points redeemed as a ratio.

After the raw analysis we found that the coordinates relative to each customer gave us the same information as the city (29 unique pairs of longitude and latitude and 29 unique cities). To extend this information we included the postal codes of the rural customers one-by-one. Some of them were fake; for the real ones we used geocoder.ca to find their location, we changed the longitude and latitude of them, and calculated the distance from the airport of the city. Other geographical insights were obtained by calculating the average travel distance for each customer. See Appendix A for the summary table.

2. External features integration

To impute certain missing data, understand potential errors, and enhance the descriptive quality of the dataset we looked for and applied information from external sources. That happened in 3 main checkpoints.

The first regarded the enhancement of the coordinates. We looked for the rural postal codes and saved their coordinates using geocoder.ca. Interpolating them with the coordinates of the airports we managed to get a more precise geographical segmentation of the customers.

The second was to check the consistency of the data of the flight length. We assessed that every customer with an average flight length greater than 13,230 km (Toronto–Manila, longest travel from Canada, FlightMapper. (n.d.)) was a problem.

The third one regarded the income for people with college education. In our case, all of our customers, whose highest level of education was a college degree, had missing values for their salaries. We have decided to use an outside source to try and impute these data points in a meaningful way. We decided to use the data from an article published by Statistics Canada (2017), where using the 2016 Census of Population data they provided the median annual salaries of Canadians by province, education level and sex. Having access to all of these attributes for our

customers we have concluded that even though this imputation would not yield a continuous scale of values, it is still stratified well enough for it to be useful for our analysis.

3. Loyalty Status

As part of our data exploration, we took a look at the already existing loyalty program, more specifically, the attributes of each loyalty group. We wanted to see whether the different loyalty ranks meant a different typical behaviour within our customer base. If some patterns emerge between the customers in the 3 levels of our program, it could be utilized in a resourceful way.
What we found, however, was the complete opposite. We looked at our existing and newly engineered features and compared their median values between the 3 levels. There seems to be little to no difference in the median values of the loyalty groups, meaning that we can not make assumptions about anyone based on their rank within our system. This implies that it would make sense to create a newer, more effective way of segmenting our customers. In the later stages of our project, that is what we are aiming to achieve using clustering algorithms.

4. Unexpected findings

By far the most interesting finding of the exploration was the difference in the annual incomes of the people with different education levels. With our imputed datapoints, the median annual incomes of people with college and bachelors degrees were more than 50000 CAD (As the database is Canadian, we assume the currency to be CAD). For the customers without a higher education degree and also the people with masters degrees and PHDs, the annual salaries were around 20000 CAD, which seems unrealistic. Maybe this has something to do with the much smaller number of observations for these categories than for the BSc and college degrees.
Another interesting observation we made is that between the provinces and territories, there is only a minimal difference in the average distance of a flight. All the averages are within a 100 km range.
The women in our dataset are very similar to the men in most of their attributes, except for their salaries. Our male customers make around 13000 CAD more annually than their female counterparts.

5. Relevant information for clustering

The main challenging aspect of our data is that a large majority of features are irrelevant when it comes to segmenting our customers. One possible explanation for this is the relative low data density of the core part of the features. In the future we are planning to continue engineering features in order to extract more information from our dataset. Another challenge is that some of our variables are highly correlated (see correlation matrix in Appendix A) which can cause the clustering to be biased. However, there are some sorts of grouping and patterns between our features, even though there are not very obvious, distinct groups. For example, in the lifetime value of our customers, in the average money spent per flight or in the length of the subscription there could be patterns that will prove to be useful for clustering.

These patterns not forming completely separate groups could indicate that fuzzy clustering is a logical method to use. The idea behind this is that with standard clustering we rely on a defined strategy to

clearly discriminate and separate groups. Since here we do not have such clear separations, we assume that every customer belongs to all groups to a certain degree. Analyzing these participation percentages can help us gain insights into customers who sit more in between two groups rather than clearly belonging to just one. We can also conclude that a smaller participation in a certain cluster can sometimes be more impactful than a larger one.

Before the actual clustering process it is not entirely clear how many groups we will end up having, but we are aiming to create 3-5 groups, to create an easy to understand but also useful method of segmenting our customers.

5. Conclusion

In this project, we completed the first critical step in building a data-driven customer segmentation strategy for AIAI. Starting with two raw datasets, we addressed data quality issues, addressed missing values, and enriched key information such as income, geography, and travel distances. This cleansing and feature engineering effort allowed us to transform messy data into a structured and reliable foundation for subsequent analysis. This exploration revealed some significant insights: current loyalty program levels do not correspond to real differences in customer behavior; certain distributions (such as income by education level) suggest anomalies or unexpected dynamics; and geographic differences in the customer data are very limited due to high clustering in metropolitan areas. We also identified a limited set of truly useful variables for distinguishing customers and detected signs of structure in behavior, although not yet clearly separated groups. This suggests that flexible clustering approaches (K-Means, DBSCAN), as well as fuzzy clustering, may be the best solution for the given data. Furthermore, before clustering we are planning on experimenting with additional features such as recency, frequency, monetary values, that could contribute to the segmentation of our customers. It is also to be decided which customers we want to include in the clustering process, for example whether we want to keep the people who cancelled their subscriptions or not. As a result of this analytical roadmap we expect to obtain 3–5 meaningful and actionable customer clusters. Ultimately, paving the way for a more effective customer engagement strategy for AIAI.

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Appendix A

Table: 1 Summary statistics for raw CustomerDB

Column Name	Meaning	Missing values
Loyalty#	Unique identifier for members	-
First Name	Customer's first name	-
Last Name	Customer's last name	-
Customer Name	Customer's full name	-
Country	Customer's country of residence	-
Province or State	Customer's province or state of residence	-
City	Customer's city of residence	-
Latitude	Geographic latitude of the city	-
Longitude	Geographic longitude of the city	-
Postal code	Customer ZIP code	-
Gender	Customer's gender	-
Education	Customer's education level	-
Location Code	Customer's residence classification	-
Income	Customer's annual income	20
Marital Status	Customer's marital status	-
Loyalty Status	Current loyalty program	-
Enrollment Date Opening	Joining date of the program	-
Cancellation Date	Date when the customer left the program	14327
Customer Lifetime Value	Monetary value of the customer	20
Enrollment Type	How they found the program	-

Table 2: Summary statistics for raw FlightsDB

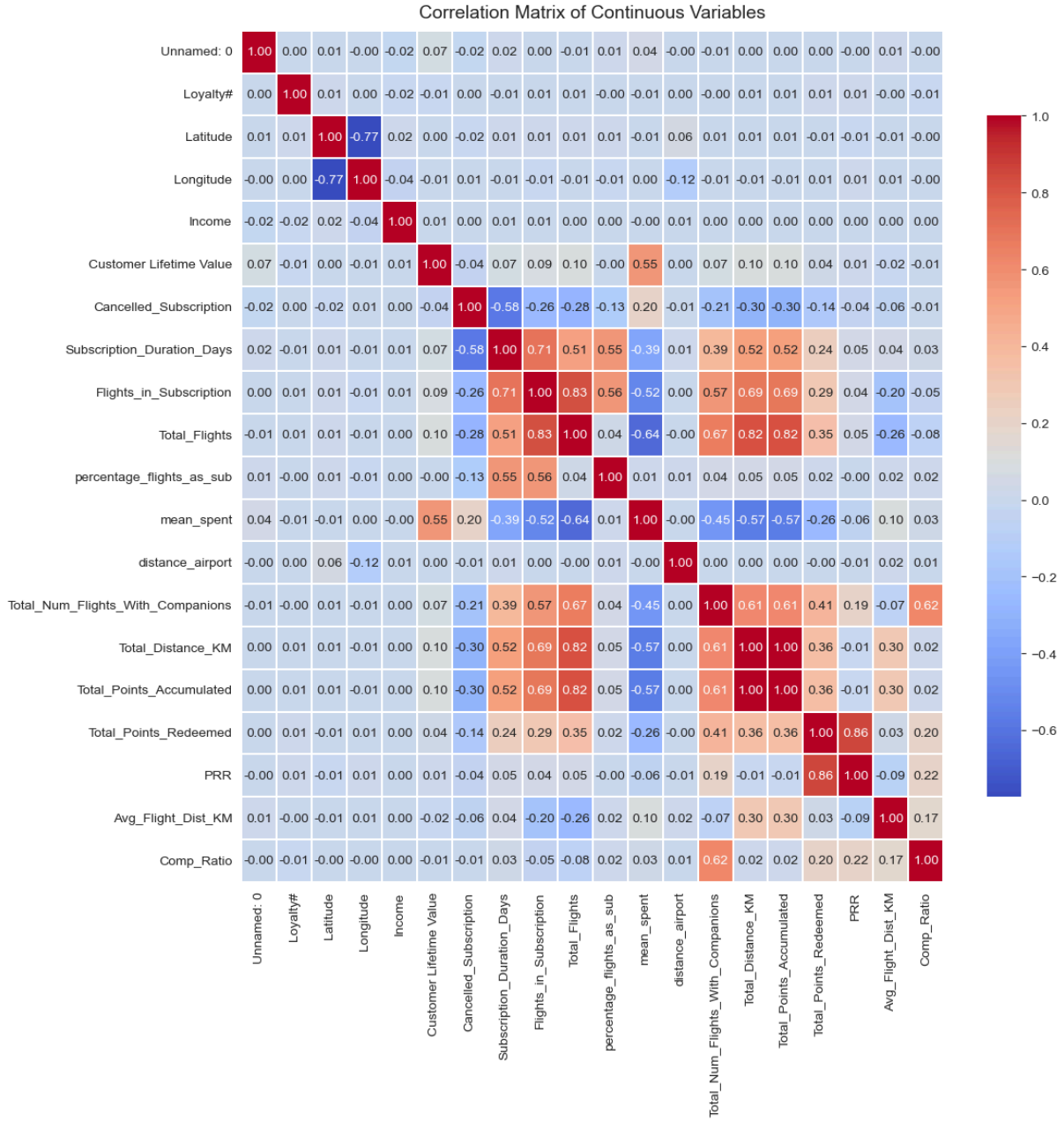
Column Name	Meaning	Missing values
Loyalty#	Unique identifier for members	-

Year	Year of the recorded flight activity	-
Month	Month of the recorded flight activity	-
Year Month Date	First day of the month with recorded activity	-
Flights	Number of flights taken in the month by the customer	-
Flights with companions	Number of flights taken in the month by the customer with companions	-
Distance KM	Distance travelled in the mont	-
Points accumulated	Loyalty points earned by the customer in the month	-
Points redeemed	Loyalty points redeemed in the mont	-
Dollar cost points redeemed	Dollar value of the points redeemed in the month	-

Table 3: Summary table for engineered features

Column	Meaning
Subscription duration	Duration of the subscription in days, cancelled or not
Flight in subscription period	How many flights the customer took as a subscriber
Percentage of flights in the subscription	The percentage of flights taken by the customer during the subscription
Companion flights ratio	Percentage of flights taken with companions
Average money spent for flight	How much a customer spent on average for a flight
Total points earned	Cumulative sum of the points earned by the customer
Total points redeemed	Cumulative sum of the points redeemed by the customer
Point's redemption rate	Percentage of points redeemed by the customer
Distance from airport	How far the customer lives from the airport (rural customers)
Average travel distance	Average flight distance for every customer

Figure 1 Correlation matrix



ANNEXES

Annex AI Usage Statement

Claude: Brainstorm visualization ideas and example syntax generation

Report Proofreading (Grammarly for grammar checks)

ChatGPT: Troubleshooting the code and helping with visualization; Help with map design, choose colour and right formatting of the EDA dashboard

All analytical insights, business interpretations, and strategic recommendations represent original group analysis and thinking.

Annex Contribution Statement

Moritz Baldauf (Student ID: 20250370)

- Creation of Infographic poster
- Introduction section in the Report

Libero Biagi (Student ID: 20250349)

- Feature understanding and cleaning
- Partial feature engineering
- Data analysis section in the report

Jenny Cubelo, (Student ID: 20250431)

- EDA dashboard

Marcel Hajdú, (Student ID: 20250377)

- Helping with data cleaning
- Feature engineering
- Visualization of pre-processed data
- Results paragraph in the report

All members contributed to collaborative discussions and ideation.

Annex Responsibility Statement

We, the group members listed above, certify that this report represents our original analytical work and interpretations. While AI tools were used as specified above, all insights, conclusions, and recommendations are the result of our independent analysis and critical thinking. We take full responsibility for the accuracy and quality of this submission

Annex Interactive EDA Dashboard

This Interactive dashboard provides an Exploratory Data Analysis (EDA) of the AIAI Customer Loyalty dataset. It works on a dynamic filtering system where all components are interconnected by a filter segmentation panel. It enables stakeholders to explore immediately how customer attributes impact key metrics, geographical and loyalty distribution.

Filter Segmentation side bar

The Filter segmentation side bar is the control center of the dashboard, any selection or removal of filters updates instantly all the present elements. It's a multidimensional filtering across three categories:
Province/State, Educational Level and Loyalty Status.

Key Indicators

They provide an immediate overview of filtered segments, helping stakeholders to quickly visualize and assess size and characteristics: Total Customers, Avg Customer Lifetime Value, Avg Flight Distance and Avg. PRR

Interactive geographic Map

It's a scatter plot over a North American map where it shows customer distribution. Points are color coded by the average flight distance and the size of the bubble indicates the total of flights.
This way the user can easily identify geographically patterns and differences.

Loyalty Status pie chart

The chart allows users to visualize the percentage of how customers are distributed across the three loyalty statuses: Star, Nova, Aurora. It makes it easy to identify which loyalty status has more or less members.

Instructions on How to run the dashboard

1. Install streamlit in your environment
    - In the terminal run: pip install dashboard.py
    - In the Anaconda environment, open Anaconda prompt and type:
      conda install -c conda-forge streamlit

2. Run notebook data_mining.ipynb, this will generate the dashboard.py file in the project folder
3. Navigate to the project folder in terminal
4. Run the dashboard with the command: streamlit run dashboard.py
5. The dashboard will automatically open in a new tab in your default browser